B2.2 Organelles and compartmentalization

NAME

DATE

How to use this sheet: go through the slides with it, then close them and write from memory. Use your own words and add page numbers from the book (pp. 244–247) where you want more detail.

Guiding questions. How does the structure of each organelle suit its function? Why is it useful for cells to have separate compartments?

B2.2.1 Organelles

Define organelle. Then sort the structures below.

TWO MEMBRANES	ONE MEMBRANE	NO MEMBRANE	NOT ORGANELLES

B2.2.2 Nucleus and cytoplasm

	EUKARYOTE	PROKARYOTE
Where the DNA is		
Timing of translation		

What advantage does this separation give eukaryotes?

B2.2.3 Compartments in the cytoplasm

List four advantages of compartments.

Explain how lysosomes and phagocytic vacuoles show these advantages.

Before you finish

Tick the ones you can explain without notes. Circle anything to ask about.

- ☐ Definition of organelle
- ☐ Sorting structures
- ☐ Nuclear membrane
- ☐ Modifying mRNA
- ☐ Concentrating enzymes
- ☐ Lysosomes
- ☐ Phagocytic vacuoles